

CoLD: Counterfactually-Guided Length Debiasing for Process Reward Models in Mathematical Reasoning

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Abstract

Process Reward Models (PRMs) play a central role in evaluating and guiding multi-step reasoning in large language models (LLMs), especially for mathematical problem solving. However, we identify a pervasive length bias in existing PRMs: a tendency to assign higher scores to more verbose reasoning steps, regardless of their semantic content or logical validity. This bias undermines the reliability of reward predictions and leads to overly verbose outputs during inference. To address this issue, we propose **CoLD (Counterfactually-Guided Length Debiasing)**, a unified framework that mitigates length bias based on counterfactual reasoning and causal graph analysis through three components: (1) an explicit length-penalty module, (2) a trainable bias estimator to capture spurious length-related signals, and (3) a joint training strategy that disentangles semantic correctness from superficial length features. Extensive experiments on MATH500 and GSM-Plus show that CoLD improves accuracy in step selection, and encourages more concise, logically valid reasoning. Furthermore, it consistently improves downstream RL performance and generalizes across domains by mitigating length bias, demonstrating CoLD’s strong generalization capability.

1 Introduction

Large language models (LLMs) have shown strong mathematical reasoning ability [26, 8, 58, 31, 20, 42], yet their solutions often contain hidden reasoning errors despite yielding correct final answers [19]. Process Reward Models (PRMs) [19, 38, 54] were introduced to evaluate the logical soundness of intermediate steps and are now widely used in both error diagnosis and inference-time scaling strategies [40, 34, 52]. Despite the impressive performance of recent PRMs [57, 36, 25, 59], we observe that their reward predictions can be unduly influenced by superficial features of reasoning steps, particularly their step-level textual length. We term this phenomenon **length bias**: *the tendency of PRMs to assign higher scores to textually longer or more verbose steps, even if their semantic content and logical correctness are identical to more concise counterparts*. Such bias undermines the reliability of PRMs, as it confuses surface-level verbosity with the quality of substantive reasoning.

To investigate this, we designed a controlled experiment decoupling step length from reasoning quality. Based on existing datasets [18], we constructed a semi-synthetic dataset where steps are extended via duplication or verbose LLM rewriting, strictly preserving their original semantics and correctness. We then input both the original and extended versions into the same PRM, recording the reward scores and token lengths. As shown in Figure 1, the textually longer steps consistently

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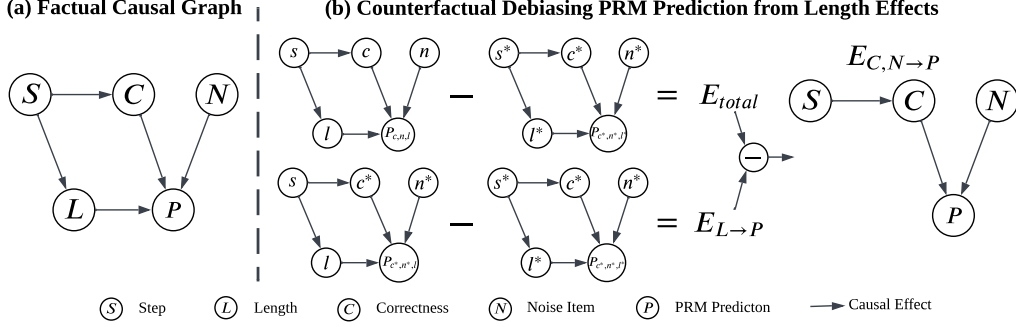


Figure 2: Causal Modeling to Eliminate Length Bias in Prediction with Causal Graph of Factors Affecting PRM Reward. Uppercase letters (e.g., L) represent random variables, lowercase letters (e.g., l) denote specific values, and lowercase letters with an asterisk (e.g., l^*) indicate fixed values. E denotes the causal effect.

receive higher rewards despite being logically equivalent to the originals. This suggests that PRMs exploit step length as a spurious shortcut, inflating scores in ways that undermine their reliability in assessing true reasoning quality.

To further analyze the factors influencing PRM predictions, we construct a causal graph [28] that captures the qualitative relationships among key variables, as illustrated in Figure 2(a). The graph includes five nodes: the input step (S), length (L), logical correctness (C), latent noise factors (N), and PRM prediction (P). Ideally, predictions should follow the path $S \rightarrow C \rightarrow P$, ensuring that scores reflect reasoning quality. However, our analysis reveals a spurious path $S \rightarrow L \rightarrow P$, indicating that verbosity directly manipulates the reward—independent of logical value. While latent noise factors such as fluency or problem type (N) may also impact P , their influence appears secondary. Additional analysis is provided in the Preliminary section 2.2.

Addressing this issue, we adopt a counterfactual perspective. We define length bias as the undesired change in a PRM’s output when step length varies while semantics and correctness are held fixed. This perspective motivates the goal of eliminating the score component attributable solely to length, as shown in Figure 2(b). Guided by this insight, we propose **Counterfactually-Guided Length Debiasing (CoLD)**, a unified framework comprising three strategies: (1) **Length Penalty**: A simple adjustment subtracting a length-proportional term from the score; (2) **Bias Estimator**: an auxiliary model trained to estimate and remove length-induced bias to restore length invariance; (3) **Joint Training**: a unified training scheme that jointly optimizes the PRM and the Bias Estimator. By introducing input perturbations and regularization, the model is encouraged to disentangle semantic content from superficial length features.

Together, CoLD forms a principled debiasing strategy. By explicitly modeling, estimating, and removing spurious length effects, CoLD reduces the correlation between reward and verbosity while maintaining semantic fidelity. Empirically, it improves selection accuracy and promotes concise, logically sound responses.

Our main contributions are as follows:

- We are the first to identify and empirically validate length bias in PRMs, showing the tendency of PRMs to favor verbose steps over equally correct concise ones. Through causal graph analysis, we further reveal that step length is a confounding factor that distorts reward estimation and undermines the reliability of PRMs.

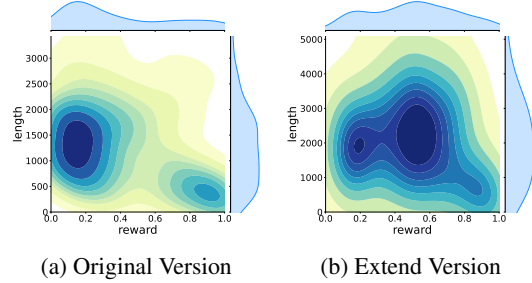


Figure 1: The joint distribution of reward score (x-axis) and step length (y-axis). (a) Distribution for the original reasoning steps. (b) Distribution for the same steps after being paraphrased by DeepSeek-V3 for increased textual length and verbosity, while maintaining semantic and logical equivalence.

- We introduce the CoLD framework, grounded in counterfactual reasoning, to mitigate length bias in PRMs. CoLD significantly improves the fairness and accuracy of reward assessments.
- Extensive experiments show that CoLD improves step-selection accuracy and reasoning conciseness, while consistently enhancing downstream RL performance and cross-domain generalization through length-bias mitigation.

2 Preliminary

2.1 Formulation

We consider the problem of assigning reward scores to intermediate reasoning steps in mathematical problem solving. Let q denote a math question and $\mathbf{s} = \{s^1, s^2, \dots, s^n\}$ be a sequence of solution steps. For each prefix $x^j = (q, s^{\leq j})$ comprising the first j steps, a PRM assigns a scalar reward:

$$r(x^j) \in (0, 1), \quad (1)$$

which is intended to reflect the semantic correctness of the partial solution up to step j .

To implement $r(x^j)$ using LLMs, we follow a scoring protocol that computes the reward based on the model’s classification between correctness and incorrectness. Specifically, given x^j , we extract the logits of two special answer tokens (e.g., '+' for correct and '-' for incorrect), and apply a softmax to compute a score:

$$r(x^j) = \frac{\exp(l_+)}{\exp(l_+) + \exp(l_-)}, \quad (2)$$

where l_+ and l_- denote the logits assigned to the positive and negative options, respectively.

2.2 Causal-driven Analysis

Causal graphs [28] provide a principled framework for modeling complex dependencies. We leverage this perspective to analyze PRM reward prediction and inform our framework design.

Figure 2(a) illustrates the causal graph underlying PRM reward prediction, each node represents a causal factor, and each directed edge $A \rightarrow B$ indicates that A exerts a direct causal influence on B .

- The causal graph consists of five nodes: S, L, C, N and P . S denotes the given problem-solving step, L represents the length of the given step, C represents the logical correctness of the problem-solving step. N represents latent noise factors such as linguistic fluency. P represents PRM’s reward prediction for this step.
- For a given step, it determines both length and correctness, resulting in causal edges $S \rightarrow L$ and $S \rightarrow C$. While L and C have no direct causal link, a confounding relation may exist. For example, extremely short steps often fail to convey valid reasoning, reducing the chance of correctness.
- Ideally, P should depend only on correctness C , i.e., $S \rightarrow C \rightarrow P$, making PRM a reliable evaluator that assigns high rewards to logically valid steps. In practice, however, the alternative path $S \rightarrow L \rightarrow P$ also plays a substantial role. Step length L directly inflates the reward (P) independent of reasoning quality. This indicates the PRM exploits verbosity as a shortcut.
- Finally, while unobserved noise N may also directly influence P , as also shown in previous work [32], its effect is comparatively minor.

To isolate reasoning quality ($S \rightarrow C \rightarrow P$) from spurious length influence ($S \rightarrow L \rightarrow P$), we formulate a counterfactual objective: ensuring rewards remain invariant to length while holding correctness fixed, as illustrated in Figure 2(b). A more detailed theoretical analysis is presented in Appendix A. Based on the above analysis, we introduce a framework to eliminate length bias.

3 Methodology

3.1 Overview of CoLD PRM

As illustrated in Figure 3, CoLD PRM guides the selection of high-quality reasoning paths by systematically eliminating length bias from step quality evaluations. Rather than a single strategy, the framework integrates three complementary methods: (1) an explicit penalty term to discourage

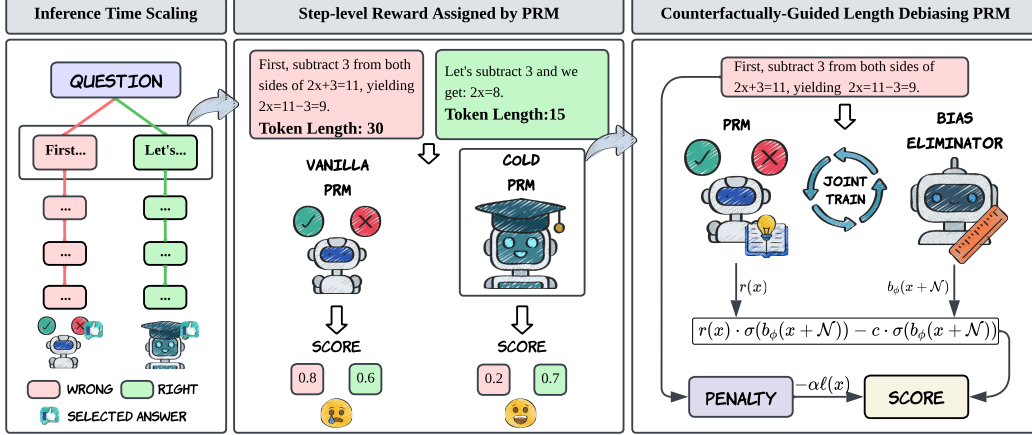


Figure 3: Overview of the Counterfactually-Guided Length Debiasing (CoLD) framework for Process Reward Models, in comparison to the vanilla PRMs.

verbosity; (2) a learnable bias estimator isolating the length-based reward component; and (3) a joint optimization scheme synchronizing the PRM and the bias estimator.

Under this framework, the final debiased reward $r^*(x)$ is computed as:

$$r^*(x) = r_\theta(x) \cdot \sigma(b_\phi(x + \mathcal{N})) - c \cdot \sigma(b_\phi(x + \mathcal{N})) - \alpha\ell(x), \quad (3)$$

where $r_\theta(x)$ denotes the score assigned by the PRM, $b_\phi(x + \mathcal{N})$ denotes the output of the bias estimator given the noise-injected input $x + \mathcal{N}$ ($\mathcal{N}$ denotes the Gaussian noise following [32]), $\sigma(\cdot)$ is the sigmoid function, c is a hyperparameter controlling the strength of bias correction, and $\alpha\ell(x)$ represents the length penalty term.

Crucially, this formulation is a direct algebraic realization of our causal counterfactual analysis (detailed in Appendix A). The core objective of counterfactual debiasing is to isolate the pure semantic effect by subtracting the bias induced by length from the total prediction. In our framework, the factual prediction $P(C = c, L = \ell)$ is estimated using the composed reward $r_\theta(x)\sigma(b_\phi(x + \mathcal{N}))$. Conversely, the counterfactual baseline $P(C = c^*, L = \ell)$, which represents the spurious score component driven purely by length when true semantic correctness is counterfactually removed, is jointly approximated by both the neural estimator $\sigma(b_\phi(x + \mathcal{N}))$ and the structural prior $\alpha\ell(x)$. By executing this explicit subtraction during inference, CoLD successfully neutralizes the spurious $S \rightarrow L \rightarrow P$ pathway and recovers the debiased semantic reward.

Further analytical insights into this formulation are provided in the following discussion.

3.2 Length Penalty

While neural networks are highly capable of modeling bias, they can exhibit unstable predictions or saturation when encountering out-of-distribution, excessively long reasoning steps. To provide a robust safeguard, we first introduce the Length Penalty item $\alpha\ell(x)$. This term serves as a deterministic structural prior that explicitly penalizes verbosity by subtracting a length-proportional term from the original PRM score.

Here, $\alpha > 0$ is a hyperparameter controlling the penalty strength, and $\ell(x)$ denotes the token-level length of the reasoning step x . This formulation introduces an explicit dependency on length into the reward computation, encouraging concise responses and discouraging unnecessary verbosity.

3.3 Bias Estimator

Length bias in PRMs is driven not merely by absolute token counts, but also by implicit, non-linear stylistic features (e.g., redundant phrasing or verbose transitions) that spuriously correlate with high rewards. To capture these complex verbosity patterns, we introduce a learnable module $b_\phi(x)$, designated as the Bias Estimator. The primary goal of the Bias Estimator is to flexibly model and

eliminate the spurious reward components attributable to these verbose styles, thereby isolating superficial bias from the true semantic reward.

To prevent the estimator from absorbing actual reasoning semantics, we apply feature-space noise perturbation. Specifically, the input is defined as $\hat{x} = x + \mathcal{N}$, where $\mathcal{N}$ denotes injected Gaussian noise [32] (Implementation details and analysis are in Appendix F). This design discourages b_ϕ from relying on semantic content and guides it to concentrate on surface-level factors such as length.

Together, the Bias Estimator and the Length Penalty form a complementary defense: the former targets non-linear features, while the latter enforces a reliable global boundary on absolute length.

3.4 Joint Training of PRM with Bias Estimator

Furthermore, we adopt a joint training strategy to simultaneously optimize the PRM $r_\theta(x)$ and the Bias Estimator $b_\phi(x)$. To disentangle correctness from stylistic features (e.g., length), we incorporate complementary correlation constraints.

We enforce complementary correlation constraints to ensure effective disentanglement between semantic correctness and length. Specifically, we measure the Pearson correlation between each module’s output and the step length $\ell(x)$:

$$\rho_r = \frac{\text{Cov}(r_\theta(x), \ell)}{\sigma_r \sigma_\ell} = \frac{\mathbb{E}[(r_\theta(x) - \mathbb{E}[r_\theta(x)])(\ell - \mathbb{E}[\ell])]}{\sigma_r \sigma_\ell}, \quad (4)$$

$$\rho_b = \frac{\text{Cov}(b_\phi(x), \ell)}{\sigma_b \sigma_\ell} = \frac{\mathbb{E}[(b_\phi(x) - \mathbb{E}[b_\phi(x)])(\ell - \mathbb{E}[\ell])]}{\sigma_b \sigma_\ell}, \quad (5)$$

where $\text{Cov}(\cdot, \cdot)$ denotes empirical covariance, and σ_r , σ_b , and σ_ℓ are the standard deviations. A small ρ_r indicates that the PRM has reduced its reliance on length, while a large ρ_b confirms that the Bias Estimator has effectively captured the length-dependent component. The corresponding correlation losses are defined as:

$$\mathcal{L}_{\text{PRM}}(\theta) = \lambda_r \cdot \rho_r^2, \quad (6)$$

$$\mathcal{L}_{\text{Bias}}(\phi) = -\lambda_b \cdot \rho_b^2, \quad (7)$$

where λ_r and λ_b control the strength of regularization. This asymmetric formulation promotes semantic fidelity in the PRM, while allowing the Bias Estimator to isolate and model spurious stylistic factors such as length.

However, relying solely on correlation-based constraints may lead to degenerate solutions that overlook semantic correctness. To ensure the debiased reward can still distinguish correct from incorrect steps, we add a cross-entropy loss as an additional supervision signal.

The composed reward is defined as $\hat{r}(x) = r_\theta(x) \cdot \sigma(b_\phi(\hat{x}))$, and is trained using correctness labels $y \in \{0, 1\}$ through a cross-entropy objective:

$$\mathcal{L}_{\text{CE}}(\theta, \phi) = -\mathbb{E}_{(x, y) \sim \mathcal{D}} \left[y \log \sigma(\hat{r}(x)) + (1 - y) \log(1 - \sigma(\hat{r}(x))) \right], \quad (8)$$

where $\mathcal{D}$ denotes the training data distribution. This loss encourages the composed reward to align with the ground-truth correctness signal.

The final training objective combines all components:

$$\mathcal{L}_{\text{Final}}(\theta, \phi) = \mathcal{L}_{\text{CE}}(\theta, \phi) + \mathcal{L}_{\text{PRM}}(\theta) + \mathcal{L}_{\text{Bias}}(\phi). \quad (9)$$

Why training and inference use different formulations. This design addresses a fundamental challenge in the field of causal inference: the absence of ground-truth labels for counterfactual data. In real-world training datasets, the observed correctness labels y are influenced by the confounding variable of inherent "verbosity," meaning they strictly belong to the observational factual data. Practically, we can’t observe a counterfactual version of the same reasoning step where its correctness is removed while its surface length is held unchanged. Consequently, the counterfactual baseline required by the theoretical formula is completely unattainable during the training phase.

Therefore, our strategy is to separate learning from debiasing. During the training stage, we deliberately retain the biased structure within the composed reward $\hat{r}(x)$ so that it completely fits the factual

Table 1: Performance comparison between CoLD and baseline RL debiasing methods. Results are reported as best-of-16 accuracy. The best and second-best results are highlighted in **bold** and underlined, respectively. * denotes significant improvement over baseline (paired t-test, $p < 0.05$)

Policy Model	Debias Method	MATH500		GSM-Plus		Avg	
		ArithACC(%)	Length	ArithACC(%)	Length	ArithACC(%)	Length
Llama-3-70B-Instruct	Vanilla-Base-PRM	44.8	555.7	71.0	324.6	57.9	440.1
	+ Length Penalty	44.8	495.1	71.5	300.1	58.2	397.6
	+ Loose Lips Sink Ships	45.8	435.7	<u>72.5</u>	302.5	59.2	368.9
	+ Adaptive Length Bias Mitigation	43.0	594.8	72.2	364.3	57.6	479.6
	+ Uniform Average	40.4	608.5	70.4	324.6	55.4	466.6
	+ Locally Weighted Regression	39.0	590.7	69.6	299.9	47.2	445.3
	+ CoLD(w/o Joint) (Ours)	<u>48.0*</u>	<u>370.4*</u>	72.1	<u>258.6*</u>	<u>60.1*</u>	<u>314.5*</u>
	+ CoLD (Ours)	49.2*	313.2*	73.8*	202.5*	61.5*	257.9*
MetaMath-Mistral-7B	Vanilla-Base-PRM	37.0	555.2	59.5	335.5	48.2	445.4
	+ Length Penalty	37.6	448.8	59.1	313.9	48.4	381.4
	+ Loose Lips Sink Ships	34.5	441.2	58.0	340.0	46.3	390.6
	+ Adaptive Length Bias Mitigation	31.0	610.0	56.5	385.0	43.8	497.5
	+ Uniform Average	33.2	609.9	56.3	363.8	44.8	486.9
	+ Locally Weighted Regression	29.6	507.3	52.7	354.4	41.2	430.9
	+ CoLD(w/o Joint) (Ours)	38.6*	353.4*	<u>59.8</u>	<u>262.7*</u>	<u>49.2*</u>	<u>308.1*</u>
	+ CoLD (Ours)	<u>37.2</u>	<u>376.3*</u>	61.4*	238.6*	49.3*	307.5*
Muggle-Math-13B	Vanilla-Base-PRM	30.4	411.1	59.1	287.9	44.8	349.5
	+ Length Penalty	30.0	376.1	59.3	280.5	44.7	328.3
	+ Loose Lips Sink Ships	28.6	375.7	59.1	263.9	43.9	319.8
	+ Adaptive Length Bias Mitigation	25.4	472.0	56.3	364.5	40.9	418.3
	+ Uniform Average	27.4	440.8	55.3	328.7	41.4	384.8
	+ Locally Weighted Regression	23.5	470.5	50.7	312.3	37.1	391.4
	+ CoLD(w/o Joint) (Ours)	<u>31.0*</u>	<u>329.0*</u>	<u>59.9*</u>	238.3*	<u>45.5*</u>	<u>283.7*</u>
	+ CoLD (Ours)	31.4*	309.2*	60.3*	<u>243.3*</u>	45.9*	276.3*

distribution $P(C = c, L = \ell, N = n)$ that inherently contains confounding factors. Built upon this, the joint optimization framework ensures that the internal sub-networks accurately partition and model this biased signal. In the inference stage, we then apply Eq. 3 to perform an explicit counterfactual subtraction in a purely algebraic manner, thereby deriving a length-debiased reward that isolates semantic quality from superficial verbosity.

Further explanation of the algorithm is provided in Appendix B.

Alternative Strategy. In addition to joint training, an alternative strategy is to train the Bias Estimator independently while keeping the PRM fixed. This approach offers greater flexibility and modularity, especially when modifying existing PRMs without re-training the whole model.

4 Experiments

In this section, we present the experimental settings and results. Our code implementation is available in the repository³.

4.1 Experiment Setup

Datasets and Metrics Our models are trained on the human-annotated PRM800K [19] and the auto-labeled Math-Shepherd [38], adopting established evaluation protocols to assess model.

For evaluation, we use the BON@n metric [19, 38] to assess the PRM’s ability to select the most correct reasoning trajectory from n candidates. For each question, the PRM assigns scores to individual steps, and the overall score of a trajectory is determined by the minimum score among its constituent steps, following prior work [38]. Details about datasets can be found in Appendix C.1.

Baselines and Implementation Details We consider a range of baselines and base models. Additional details, including further introductions to the baselines and base models, as well as hyperparameters and training sizes, are provided in Appendix C.1.

³<https://anonymous.4open.science/r/CoLD-PRM-B640/>

Table 2: Generalization of CoLD using the Llama-3-70B-Instruct. Performance is evaluated using Best-of-16 accuracy. $\uparrow$ / $\downarrow$ indicate increase/decrease in accuracy/length. * denotes significant improvement over baseline (paired t-test, $p < 0.05$). Additional results are available in Table 4, 5.

PRM Model	MATH500		GSM-Plus		Avg	
	ArithACC(%)	Length	ArithACC(%)	Length	ArithACC(%)	Length
Vanilla-Math-Shepherd-PRM	46.4	458.9	72.3	268.7	59.4	363.8
CoLD(w/o Joint)-Math-Shepherd-PRM	47.2 $\uparrow$ *	332.4 $\downarrow$ *	72.5 $\uparrow$	258.0 $\downarrow$ *	59.9 $\uparrow$	295.2 $\downarrow$ *
Vanilla-Qwen2.5-Math-PRM	45.0	595.2	73.9	284.3	59.5	439.8
CoLD(w/o Joint)-Qwen2.5-Math-PRM	49.0 $\uparrow$ *	330.6 $\downarrow$ *	74.4 $\uparrow$ *	217.6 $\downarrow$ *	61.7 $\uparrow$ *	274.1 $\downarrow$
Vanilla-EurusPRM-Stage1	49.0	546.6	73.5	308.2	61.3	427.4
CoLD(w/o Joint)-EurusPRM-Stage1	50.6 $\uparrow$ *	282.1 $\downarrow$ *	73.8 $\uparrow$	237.7 $\downarrow$ *	62.2 $\uparrow$ *	259.9 $\downarrow$ *

Table 3: Ablation study of CoLD PRM variants using Best-of-16 search. Solutions are sampled from Llama-3-70B-Instruct. We evaluate the contribution of each module through component-wise ablations. Additional results are available in Table 6 and Table 7.

Components			MATH500		GSM-Plus		Avg	
Joint Train	Bias Estimator	Penalty	ArithACC(%)	Length	ArithACC(%)	Length	ArithACC(%)	Length
$\checkmark$	$\checkmark$	$\checkmark$	49.2	313.2	73.8	202.5	61.5	257.9
$\checkmark$	$\checkmark$	$\times$	<u>48.2</u>	494.9	<u>73.2</u>	<u>237.4</u>	<u>60.7</u>	366.2
$\times$	$\checkmark$	$\checkmark$	48.0	<u>370.4</u>	72.1	258.6	60.1	<u>314.5</u>
$\times$	$\checkmark$	$\times$	44.8	525.0	71.5	303.6	58.2	414.3
$\times$	$\times$	$\checkmark$	43.2	495.1	71.5	300.1	58.2	397.6
$\times$	$\times$	$\times$	44.8	555.7	71.0	324.6	57.9	440.2

4.2 Overall Performance

We present the performance of CoLD PRM, along with various baselines and base models, across two evaluation datasets: MATH500 [12] and GSM-Plus [17]. The main results are provided in Tables 1 and 2, and the key findings are outlined as follows:

- **CoLD PRM achieves state-of-the-art accuracy while favoring significantly shorter steps, effectively mitigating length bias.** This effect is particularly pronounced on MATH500, where trajectories are more complex and prone to verbosity. By correcting for length-induced distortions, CoLD PRM demonstrate its robustness and superior evaluative performance in challenging scenarios.
- As shown in Table 1, **while these baselines effectively address reward bias in RL, they struggle to generalize to length bias in PRMs**, yielding suboptimal accuracy and verbose responses. In contrast, our CoLD framework significantly improves both predictive accuracy and response conciseness.
- Table 2 demonstrates that even **without joint training, combining a bias estimator with length-penalty correction substantially improves performance across various base models**. This highlights the effectiveness of these components and suggests that our framework can be flexibly applied to existing, pre-trained PRMs. In settings with limited computational resources, this provides a practical and efficient way to improve reward quality without retraining the entire model.
- It is worth noting that **shorter length does not inherently equate to higher accuracy**. For instance, Math-Shepherd-PRM selects shorter reasoning steps yet exhibits lower accuracy. A model that aggressively favors brevity may overlook essential reasoning steps. The goal of CoLD is not to shorten responses indiscriminately, but rather to reduce unnecessary verbosity while preserving or even improving accuracy.

4.3 Ablation Study

To understand the contribution of each component in our CoLD PRM framework, we conduct a comprehensive ablation study by selectively removing the Joint Training, Bias Estimator, and Length Penalty variants. The performance of these variants is presented in Table 3, from which we can draw the following observations:

- **Removing Joint Training leads to a noticeable increase in the average solution length across both datasets**, indicating that without joint optimization, the reward model tends to favor longer reasoning trajectories. This suggests that joint training helps align the reward signal with downstream policy behavior, implicitly regularizing verbosity.

- **Excluding the Bias Estimator similarly leads to longer solutions, while the accuracy remains relatively stable.** This points to the Bias Estimator’s role in explicitly identifying and correcting for length-related confounding effects in the reward scores, encouraging the model to recognize correctness independent of surface-level length cues.
- **Using the Length Penalty alone reduces length but fails to improve correctness, and can even lead to a degradation in overall accuracy.** This suggests that a naive penalty on length, when not informed by bias estimation, may remove beneficial reasoning steps along with redundant ones, highlighting the risk of applying heuristic penalties without semantic guidance.

4.4 Debias Visualization

To better understand the effect of our debiasing framework, we visualize the reward distributions after applying CoLD. As discussed in the introduction and shown in Figure 1, the reward model favors longer reasoning steps even when they are semantically equivalent to shorter ones.

After applying CoLD, we observe a marked improvement in the alignment between the rewards assigned to original and extended steps. As shown in Figure 4, the distributions of scores become substantially more consistent, indicating that the model no longer favors more verbose steps. This suggests that CoLD PRM successfully removes the spurious correlation between step length and reward.

Notably, the debiased model places greater emphasis on the semantic correctness and logical validity of each step, rather than its superficial length. This demonstrates that our method not only improves quantitative performance but also corrects length bias in a principled manner.

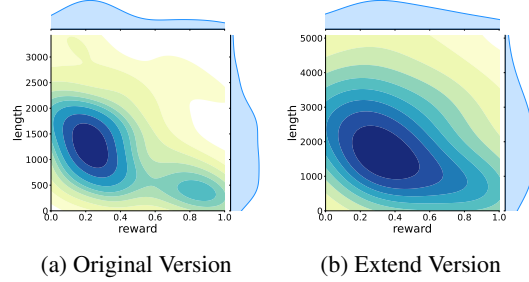


Figure 4: The joint distribution of rewards and step lengths after debiasing on both original and length-augmented steps.

4.5 Downstream RL application

To demonstrate CoLD’s effectiveness on downstream tasks, we apply CoLD in a reinforcement learning setting. Specifically, our experiments are conducted within the ReasonFlux-PRM framework [59]. We use Qwen2.5-Math-1.5B-Instruct as the policy model and adopt GRPO (Group Relative Policy Optimization) [31] as the RL optimization algorithm. CoLD is integrated into the ReasonFlux-PRM and compared against its vanilla counterpart. For evaluation, we use MATH500 [19] as the test benchmark.

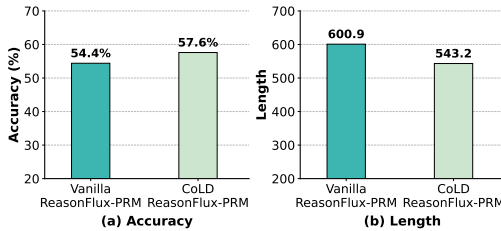


Figure 5: Performance of PRMs as reward signals in policy optimization.

As shown in the Figure 5, incorporating CoLD not only improves the accuracy of the policy model in problem solving, but also reduces the output length. This indicates that CoLD effectively mitigates the length bias exhibited by the PRM, **providing the model with more precise and meaningful reward signals during downstream reinforcement learning.**

4.6 Cross-domain Generalization Study

To evaluate whether CoLD can generalize to domains beyond its original scope, we integrate it into VersaPRM [47], a process reward model designed to operate across heterogeneous domains. We assess its performance on the MMLU-Pro [39] benchmark, a robust and challenging massive multi-task understanding dataset created to rigorously test the capabilities of large language models. CoLD

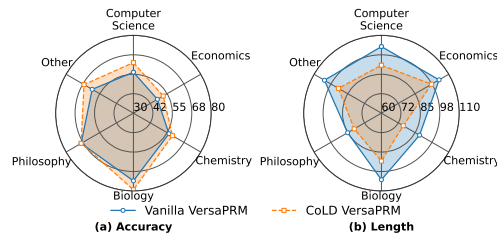


Figure 6: Generalization performance of CoLD PRM on cross-domain tasks.

is applied within VersaPRM across several representative domains, including Computer Science, Economics, Chemistry, Biology, Philosophy, and Other.

Based on the results shown in the Figure 6, when applying CoLD on top of VersaPRM, we observe consistent improvements in both accuracy and the reduction of response length across these heterogeneous domains. This demonstrates that the debiasing principle behind CoLD does not rely on domain-specific structures found in mathematical reasoning, but instead **addresses a general phenomenon of superficial-feature bias in step-level reward modeling**.

4.7 Additional Experiments

- **Scaling Width Study** In Appendix E, We evaluate CoLD across varying scaling width($N \in \{2, 4, 8, 16\}$), demonstrating that CoLD consistently outperforms vanilla PRMs.
- **Hyperparameter Study** A detailed analysis of hyperparameters c , λ_b , and α is provided in Appendix G, highlighting CoLD’s inherent parameter robustness.
- **Other Benchmark** In Appendix H, we evaluate CoLD on the representative ProcessBench benchmark. The results demonstrate that CoLD successfully mitigates bias while preserving the PRM’s core evaluative capabilities.
- **Evaluation Using Stronger Policy Model** Appendix I provides additional evaluations using Qwen3-Next-80B-A3B-Instruct [42] and DeepSeek-v3.1 [20], where CoLD exhibits robust performance across these advanced model.
- **Bias Estimator Interpretability** We conduct interpretability analyses focusing on its sensitivity to input length versus semantic content in Appendix J.

5 Related Works

5.1 Process Reward Models

Process Reward Models (PRMs) enable fine-grained step-level supervision for model reasoning, addressing traditional Outcome Reward Models (ORMs) [6] limitation of only scoring final outputs. This design mitigates "spurious correctness" and boosts stability in complex tasks. A key challenge for PRM training is high annotation costs: PRM800K [19] was the first human-annotated dataset, while later works [37, 23, 51] used LLMs and Monte Carlo (MC) methods to reduce costs. PRMs have advanced for both RL optimization [29, 3, 36, 4, 50, 11] and inference time scaling [24, 41, 52, 30, 14, 13]. Despite these advances, few PRM-related works have focused on the length bias issue inherent in PRMs—and this is precisely the problem addressed by our proposed CoLD PRM.

5.2 Causal Debias

Causal debias has emerged as a fundamental strategy across diverse machine learning domains [56, 45, 35]. For example, Zhang et al. [49] mitigates bias in multi-hop fact verification by applying front-door adjustment to reasoning paths and estimating causal effects via random walks. Chisca et al. [5] reduce LLM bias by leveraging causal paths to design prompts that emphasize factual knowledge over stereotypes. Zhou et al. [55] propose a fine-tuning framework that integrates causal intervention and invariant risk minimization to suppress reliance on non-causal, bias-inducing factors. Liu et al. [22] introduces a counterfactual-enhanced framework that debiases multimodal sentiment classification through adaptive contrastive learning. Liu et al. [21] propose a causal framework with data augmentation to filter out irrelevant artifacts. Zhan et al. [48] presents a counterfactual training strategy for Med-VQA, removing spurious linguistic correlations via causal intervention. Farzam and Sapiro [10] analyzes how differential privacy can bias causal estimates and proposes robust regression techniques to correct such distortions. While these methods effectively address domain-specific confounders, none directly tackle the unique length bias [27, 9] in process reward models (PRMs). Our proposed CoLD PRM fills this gap by disentangling semantic correctness from spurious correlations with output length.

6 Conclusion

In this work, we identify and systematically study length bias in process reward models (PRMs). To tackle this issue, We introduce CoLD, a Counterfactually-Guided Length Debiasing framework. Specifically, CoLD incorporates a length penalty, a bias estimator, and joint training to mitigate the undue influence of response length on reward scores. Extensive experiments show that CoLD improves solution-selection accuracy and RL performance by mitigating length bias, establishing a generalizable and robust strategy for process reward modeling.

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A Causal Counterfactual Analysis

In the causal graph shown in Figure 2, variables influence one another. For example, both correctness C and step length L affect the prediction P . Therefore, the value of P can be determined by its ancestor nodes, mathematically expressed as:

$$P_{c,n,l} = P(C = c, N = n, L = \ell). \quad (10)$$

where $P(\cdot)$ denotes the value function associated with P .

To rigorously quantify the effect of these variables on the model prediction, we adopt counterfactual methods. The core idea of counterfactual analysis is to evaluate the outcome under hypothetical scenarios by selectively altering certain variables. For example, we can set L to a fixed reference value ℓ^* , which represents a counterfactual intervention that removes its actual effect from the system. Since ℓ^* is held constant, it acts as a reference condition with a consistent influence on downstream variables. Likewise, we counterfactually set C to c^* to isolate the combined effect of C and L on P . Under this setting, the total effect on P can be formulated as:

$$E_{total} = P_{c,n,\ell} - P_{c^*,n^*,\ell^*} \quad (11)$$

Here, C is also replaced by c^* because c^* also has an influence on P . c^* represents the counterfactual value of P .

Moreover, based on the structure of the causal graph, we can decompose this total effect into two distinct components: (1) the effect of the correctness variable on the PRM prediction, i.e., the path $C \rightarrow P$, and (2) the effect of length, i.e., $L \rightarrow P$. To isolate the effect of length, we hold C at its counterfactual value c^* while allowing L to vary, which gives:

$$E_{L \rightarrow P} = P_{c^*,n^*,\ell} - P_{c^*,n^*,\ell^*} \quad (12)$$

where $P_{c^*,L}$ represents the prediction when the observed length L is retained, while the correctness signal C has been counterfactually removed as shown in Figure 2(b).

We perform this counterfactual removal of C because we aim to measure only the contribution of L to the prediction, independent of C . This process cannot be directly computed from observational data and thus relies on causal reasoning, making it a canonical example of counterfactual causal inference.

Once E_{total} and $E_{L \rightarrow P}$ are calculated, $E_{C,N \rightarrow P}$ can be obtained by subtracting the former from the latter, as illustrated in Figure 2:

$$\begin{aligned} E_{C,N \rightarrow P} &= E_{total} - E_{L \rightarrow P} \\ &= P_{c,n,\ell} - P_{c^*,n^*,\ell^*} - (P_{c^*,n^*,\ell} - P_{c^*,n^*,\ell^*}) \\ &= P_{c,n,\ell} - P_{c^*,n^*,\ell} \\ &= P(C = c, N = n, L = \ell) - P(C = c^*, N = n^*, L = \ell) \end{aligned} \quad (13)$$

This expression captures the isolated effect of the correctness signal C on the model prediction P , with length L held constant. As discussed in Section 3, the value $P(C = c, L = \ell)$ can be estimated using the composed reward $r_\theta(x)\sigma(b_\phi(x + \mathcal{N}))$, where $r_\theta(x)$ is the PRM score and $b_\phi(x + \mathcal{N})$ is the bias estimator output. On the other hand, $P(C = c^*, L = \ell)$ can be approximated by both $\sigma(b_\phi(x + \mathcal{N}))$ and $\alpha\ell(x)$.

Importantly, the training labels themselves inherently contain verbosity bias, and it is practically impossible to directly obtain truly unbiased labels in real-world scenarios. This is precisely why our framework is designed in this way. During training, we use the CE loss to align the composed, biased term $r_\theta(x)\sigma(b_\phi(x + \mathcal{N}))$ with the biased labels, enabling the joint model to properly fit the observational data distribution. During inference, however, we apply Eq. 3 to subtract the counterfactual bias term from the prediction. In this way, our method remains faithful to the causal interpretation: we fit the biased factual distribution during training, and then perform counterfactual intervention at inference to recover the true, debiased semantic reward.

B Algorithm Description

Algorithm 1 summarizes the joint training procedure for the Process Reward Model (PRM) and the Bias Estimator. The training aims to disentangle semantic correctness from length-related bias in the reward predictions.

At each iteration, a mini-batch of training samples is drawn from the dataset. For each sample, noise is injected into the input features before feeding them to the Bias Estimator to prevent it from capturing semantic information. The composed reward $\hat{r}(x)$ is then computed as the element-wise product of the PRM output and the sigmoid-activated Bias Estimator output.

The binary cross-entropy loss supervises the composed reward to align with ground-truth correctness labels. To encourage the PRM to focus on correctness rather than spurious length correlations, a

Algorithm 1 Joint Training of PRM and Bias Estimator

Require: Training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, learning rates η_r, η_b , hyperparameters λ_r, λ_b , bias correction factor c

- 1: Initialize PRM $r_\theta(\cdot)$ and Bias Estimator $b_\phi(\cdot)$
- 2: **while** not converged **do**
- 3: Sample mini-batch $\{(x_i, y_i)\}_{i=1}^B \sim \mathcal{D}$
- 4: **for** each x_i in batch **do**
- 5: Inject noise: $\tilde{x}_i = x_i + \mathcal{N}$
- 6: Compute $\hat{r}_i = r_\theta(x_i) \cdot \sigma(b_\phi(\tilde{x}_i))$
- 7: **end for**
- 8: Compute cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{i=1}^B [y_i \log \sigma(\hat{r}_i) + (1 - y_i) \log(1 - \sigma(\hat{r}_i))]$$

- 9: Compute Pearson correlations $\rho_r = \text{Corr}(r_\theta(x), \ell(x))$, $\rho_b = \text{Corr}(b_\phi(x), \ell(x))$
- 10: Compute module-specific losses:

$$\mathcal{L}_{\text{PRM}} = \lambda_r \cdot \rho_r^2, \quad \mathcal{L}_{\text{Bias}} = -\lambda_b \cdot \rho_b^2$$

- 11: Compute final losses:

$$\mathcal{L}_{\text{Final}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{PRM}} + \mathcal{L}_{\text{Bias}}$$

- 12: Update PRM parameters θ : $\theta \leftarrow \theta - \eta_r \nabla_\theta (\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{PRM}})$
 - 13: Update Bias Estimator parameters ϕ : $\phi \leftarrow \phi - \eta_b \nabla_\phi (\mathcal{L}_{\text{CE}} - \mathcal{L}_{\text{Bias}})$
 - 14: **end while**
-

regularization term penalizing the squared Pearson correlation between PRM predictions and step length is added. Conversely, the Bias Estimator is encouraged to maximize its correlation with step length to model the bias effectively.

Separate gradient updates are applied to the PRM and Bias Estimator parameters using their respective loss functions, allowing each module to specialize in modeling semantic correctness and length bias, respectively.

At inference time, the debiased reward is computed by correcting the PRM output with the estimated bias from the Bias Estimator, scaled by a hyperparameter controlling the bias removal magnitude.

C Experiment Details

C.1 Implementation Details

Train Dataset We use both PRM800K and Math-Shepherd as training datasets. The first, PRM800K [19], consists of 800,000 step-level correctness labels derived from the MATH dataset via extensive human annotation, offering a high-fidelity but annotation-expensive training resource. The second, Math-Shepherd [38], comprises 400,000 automatically generated labels across both MATH and GSM8K problems. It provides scalable supervision without human involvement, allowing for cost-effective training at scale.

Evaluation Dataset The evaluation datasets employed in our study are curated by Li and Li [18], drawing from the MATH500 [12] and GSM-Plus [17] datasets. The collected trajectories are sampled from three mathematical solvers of varying model scales: MetaMath-Mistral-7B [44], MuggleMath-13B [16], and Llama3-70B-Instruct [1]. In addition, we construct a semi-synthetic extension of this dataset to facilitate controlled evaluation of length-related biases. Specifically, for each original solution trajectory, we generate semantically equivalent but longer variants using two strategies: (1) duplicating individual steps to create trivially lengthened versions, and (2) prompting DeepSeek [20] to rewrite each step with increased verbosity while preserving its semantic

meaning and logical validity. These two transformation methods reflect two common patterns in model-generated outputs—verbatim repetition and verbose paraphrasing with limited substantive contribution. An example illustrating the two expansion methods can be found in Figure 7.

To further verify that LLM-based sentence expansion does not alter the original meaning (i.e., correctness), we randomly sampled 300 pairs of expanded and original sentences for manual evaluation. The results show a pass rate of 95.3%, indicating that the imposed constraints on sentence rewriting are effective and largely preserve semantic fidelity.

Based on the above, we construct a test set consisting of 16 solutions per question: 8 original trajectories and 8 length-augmented variants that retain the same underlying semantics. This design enables fine-grained analysis of model behavior under superficial variations in step length.

Hyperparameter and Training Settings For joint training, we adopt a batch size of 128, with initial learning rates of 1×10^{-4} for the PRM and 3×10^{-4} for the bias estimator. The loss weights are set as $\lambda_r = 0.1$ and $\lambda_b = 0.5$ throughout training.

For experiments that train only the bias estimator, we use a batch size of 64. When training on the Math-Shepherd PRM, the initial learning rate is set to 2×10^{-3} , with $\lambda_{\text{corr}} = 0.3$. When using our own PRM (trained on PRM800K), we set the learning rate to 2×10^{-3} , also with $\lambda_{\text{corr}} = 0.3$.

We selected Qwen-2.5-Math-7B-instruct [43] as the foundational large language model (LLM) for our PRM, and Qwen-2.5-0.5B-instruct as the base model for the bias estimator. For joint training, four NVIDIA A100 GPUs were utilized, with the training process taking approximately 5 hours to complete. While solely training the bias estimator, only one RTX 4090 GPU was required, and this training procedure lasted around 3 hours. To enhance training resource efficiency, we employed Parameter-Efficient Fine-tuning techniques LoRA. The LoRA configuration was set with a rank of 8, an alpha value of 32, and dropout set to 0.1.

Debiasing Methods in RLHF

- **Length penalty[33]:** The method adds a linear proportional penalty term to the original reward
- **Loose Lips Sink Ships[32]:** The method applies a Product-of-Experts framework that disentangles reward modeling from sequence-length effects by combining a semantic expert focused on human intent with a perturbed bias expert specialized in capturing length bias.
- **Adaptive Length Bias Mitigation[2]:** The method introduces Adaptive Length Bias Mitigation (ALBM), which disentangles length bias from the original reward and adaptively recombines the length and quality rewards based on the characteristics of each query.
- **Uniform Average[15]:** The method uses the local average reward to provide an estimation for the bias term, which can be removed, thereby approximating the true reward.
- **Locally Weighted Regression(LWR)[15]:** The method assigns weights to nearby data points, giving higher importance to those closer to the target. This ensures that proximate points significantly influence the regression. LWR then applies weighted linear regression to model the local behavior of the target function, effectively approximating the weighted average within the local context.

Basemodels

- **Math-Shepherd-PRM[37]:** generates process labels by estimating the empirical probability that a step leads to the correct answer and trains a Process Reward Model (PRM) on their published dataset.
- **Base-PRM:** trained by ourselves using the human-annotated PRM800K dataset, based on Qwen-2.5-Math-7B-instruct [43].
- **Qwen2.5-Math-PRM[51]:** adopts a two-phase data construction (data expansion, data filtering) and specific training. In expansion, it uses MC estimation with hard labels (a response is negative only if no 8 completions get the correct answer). In filtering, it uses

Table 4: Generalization of CoLD using the MetaMath-Mistral-7B policy model. Performance is evaluated using Best-of-16 accuracy. The $\uparrow$ indicates an increase in accuracy or length, while the $\downarrow$ denotes a decrease. A paired t-test was conducted to verify results. Statistically significant improvements over the baseline ($p < 0.05$) are denoted with an asterisk (*).

PRM Model	MATH500		GSM-Plus		Avg	
	ArithACC(%)	Length	ArithACC(%)	Length	ArithACC(%)	Length
Vanilla-Math-Shepherd-PRM	32.6	381.3	59.9	291.8	46.3	336.6
CoLD(w/o Joint)-Math-Shepherd-PRM	32.6	300.1 $\downarrow$ *	60.4 $\uparrow$	279.9 $\downarrow$ *	46.5 $\uparrow$	290.0 $\downarrow$ *
Vanilla-Qwen2.5-Math-PRM	36.4	648.8	62.8	292.3	49.6	470.6
CoLD(w/o Joint)-Qwen2.5-Math-PRM	39.4 $\uparrow$ *	338.4 $\downarrow$ *	62.5 $\downarrow$	235.0 $\downarrow$ *	51.0 $\uparrow$	286.7 $\downarrow$ *
Vanilla-EurusPRM-Stage1	43.2	324.0	65.4	323.6	54.3	357.7
CoLD(w/o Joint)-EurusPRM-Stage1	42.0 $\downarrow$	239.7 $\downarrow$ *	64.8 $\downarrow$	260.8 $\downarrow$ *	53.4 $\downarrow$	250.3 $\downarrow$ *

Qwen2.5-Instruct-72B as LLM-as-a-judge to verify reasoning step-by-step, and applies consensus filtering to remove instances with mismatched LLM-annotated and MC-estimated process labels for quality. For training, it uses cross-entropy loss on end-of-step tokens for binary classification.

- **EurusPRM-Stage1[7]:** is trained via Implicit PRM[46], a framework that secures free process rewards without incurring additional costs—requiring only the simple training of an ORM (Outcome Reward Model) on more affordable response-level labels. During inference, implicit process rewards are generated by performing a forward pass and computing the log-likelihood ratio at each step.
- **ReasonFlux-PRM [59]:** a trajectory-aware preference model that integrates step-level and trajectory-level supervision to provide fine-grained reward signals aligned with structured reasoning traces.
- **VersaPRM [47]:** a multi-domain PRM trained on synthetic reasoning data produced through novel data generation and annotation pipeline.

C.2 Example of Semi-synthetic Solution

We generate extended variants either by duplicating the original step or by prompting DeepSeek to produce more verbose yet semantically equivalent rewrites, as illustrated in Figure 7. The prompt used for this process is provided in Figure 8.

C.3 RL Experiment Settings

We conducted a reinforcement learning experiment using the Group Relative Policy Optimization (GRPO) [31] algorithm via the verl framework. Specifically, we fine-tuned the Qwen2.5-1.5B-Instruct [43] model. The training process utilized the `dapo_500steps` dataset with a maximum response length of 16,384 tokens to support long-form reasoning, while the GRPO group size was set to $G = 6$ per prompt. For computational efficiency, the experiment was executed on a single node with 4 GPUs, employing a learning rate of 1×10^{-6} and a KL coefficient of 0.001. We optimized memory usage by enabling gradient checkpointing, chunked prefill, and CPU parameter offloading for the reference model, while the rollout engine utilized vLLM with a tensor parallel size of 2 to maintain high throughput during the generation phase.

D Supplementary Experiments

In this section, we present additional results across different policy models, as shown in Table 4, Table 5, Table 6 and Table 7.

E Scaling Width Study

In this section, we analyze the impact of our CoLD method under varying numbers of best-of-N samples. Experiments were conducted with $N = 2, 4, 8, 16$.

Question: Convert the point $(0,3)$ in rectangular coordinates to polar coordinates. Enter your answer in the form (r,θ) , where $r > 0$ and $0 \leq \theta < 2\pi$.

Origin Version Solution:

Step 1: Figure out r . We have that $r = \sqrt{0^2 + 3^2} = 3$.

Step 2: Determine θ . The point $(0,3)$ lies on the positive y -axis, so this corresponds to an angle of $\frac{\pi}{2}$ radians (or 90 degrees). Thus, the polar coordinates are $\boxed{\left(3, \frac{\pi}{2} \right)}$.

Duplicating Version Solution:

Step 1: Figure out r . We have that $r = \sqrt{0^2 + 3^2} = 3$. Figure out r . We have that $r = \sqrt{0^2 + 3^2} = 3$.

Step 2: Determine θ . The point $(0,3)$ lies on the positive y -axis, so this corresponds to an angle of $\frac{\pi}{2}$ radians (or 90 degrees). Thus, the polar coordinates are $\boxed{\left(3, \frac{\pi}{2} \right)}$. Determine θ . The point $(0,3)$ lies on the positive y -axis, so this corresponds to an angle of $\frac{\pi}{2}$ radians (or 90 degrees). Thus, the polar coordinates are $\boxed{\left(3, \frac{\pi}{2} \right)}$.

RewriteVersion Solution:

Step 1: Determine the value of r . Given the equation $r = \sqrt{0^2 + 3^2}$, we simplify it to find that $r = 3$.

Step 2: To determine the angle θ in polar coordinates, we observe that the point $(0,3)$ is located on the positive y -axis. In the polar coordinate system, this position corresponds to an angle of $\frac{\pi}{2}$ radians (which is equivalent to 90 degrees). Therefore, the polar coordinates for this point are $\boxed{\left(3, \frac{\pi}{2} \right)}$.

Figure 7: An example of the original and extended solutions

As shown in Figure 9, our CoLD PRM consistently achieves a clear advantage across different scaling widths. For the CoLD PRM without joint training, while exhibiting slightly lower accuracy than the vanilla PRM at a few points, it outperforms the vanilla counterpart for most values of n . For the vanilla PRM, however, we observe that in some cases increasing the scaling width can paradoxically lead to a decline in accuracy despite the expectation that a larger n should provide more candidate solutions and thus improve performance. This counterintuitive result highlights the substantial impact of length bias on the model’s ability to correctly assess solution steps.

F Impact of the Noise Scale $\mathcal{N}$

In our feature-space perturbation design, the noise scale k controls the intensity of semantic corruption imposed on the underlying representation. Specifically, we inject Gaussian noise into token IDs according to the following formulation:

$$\tilde{x} = \text{clip}(x + k\epsilon, 0, V - 1), \quad (14)$$

where $\epsilon \sim \mathcal{N}(0, 1)$ is truncated to the interval $[-1, 1]$, and V denotes the vocabulary size. The final perturbed token IDs are obtained by rounding down, i.e., $\lfloor \tilde{x} \rfloor$. In our main experiments, we set $k = 5$. To systematically evaluate the rationality of this design choice and its boundary effects on model

Prompt Used to Generate More Verbose but Semantically Equivalent Rewrites
I will provide you with a multi-step math problem solution. Your task is to rewrite each step by slightly expanding it while preserving its original meaning, without over-expanding. Please keep the 'Step i:' format for each step. The number of steps in your rewritten answer must match the original—do not split or add steps. If there are any mistakes in the original solution, retain them as they are and do not correct them. Below is the given solution process: step1: ... step2:

Figure 8: Prompt Used to Generate More Verbose but Semantically Equivalent Rewrites by DeepSeek

Table 5: Generalization of CoLD using the Muggle-Math-13B policy model. Performance is evaluated using Best-of-16 accuracy. The $\uparrow$ indicates an increase in accuracy or length, while the $\downarrow$ denotes a decrease. A paired t-test was conducted to verify results. Statistically significant improvements over the baseline ($p < 0.05$) are denoted with an asterisk (*).

PRM Model	MATH500		GSM-Plus		Avg	
	ArithACC(%)	Length	ArithACC(%)	Length	ArithACC(%)	Length
Vanilla-Math-Shepherd-PRM	28.6	332.1	59.2	279.7	43.9	305.9
CoLD(w/o Joint)-Math-Shepherd-PRM	28.8 $\uparrow$	262.7 $\downarrow$ *	59.1 $\downarrow$	263.3 $\downarrow$ *	44.0 $\uparrow$	263.0 $\downarrow$ *
Vanilla-Qwen2.5-Math-PRM	30.2	399.0	60.8	251.1	45.5	325.1
CoLD(w/o Joint)-Qwen2.5-Math-PRM	31.8 $\uparrow$ *	297.1 $\downarrow$ *	60.3 $\downarrow$	222.9 $\downarrow$ *	46.1 $\uparrow$	260.0 $\downarrow$ *
Vanilla-EurusPRM-Stage1	34.0	359.1	57.6	300.6	45.8	328.4
CoLD(w/o Joint)-EurusPRM-Stage1	34.6 $\uparrow$	282.7 $\downarrow$ *	57.9 $\uparrow$	265.7 $\downarrow$ *	46.3 $\uparrow$	274.2 $\downarrow$ *

performance, we conduct a quantitative ablation study over different values of k to examine the robustness of the CoLD framework under varying noise intensities.

As shown in Table 8, when the noise injection mechanism is completely removed (i.e., $k = 0$), the model’s accuracy on both benchmarks drops noticeably, while the average length of the selected solutions increases. This phenomenon suggests that, in the absence of perturbation, the bias estimator inevitably leverages its neural approximation capacity to capture and exploit deep semantic information correlated with reasoning quality in the original feature space. Such “semantic leakage” undermines the functional decomposition in the joint optimization framework, preventing the bias estimator from being effectively constrained to focus on stripping away superficial redundant features—such as step length—that are unrelated to logical correctness. As a result, its role as a dedicated bias modeling module is compromised.

In sharp contrast, when noise is introduced with $k \in \{5, 10, 15\}$, the model exhibits high stability in both prediction accuracy and length metrics. These results collectively demonstrate that while noise injection is a crucial component for effectively characterizing and modeling length bias, the overall framework remains robust at the feature-disentanglement level and is not sensitive to moderate variations in the noise scale. Therefore, in practical deployment, this mechanism can operate reliably without requiring delicate hyperparameter tuning, thereby reducing the overall cost of usage and maintenance.

Table 6: We assess the performance of CoLD PRM variants using Best-of-16 search with solutions sampled from the Muggle-Math-13B model. Component-wise ablations are conducted to evaluate the contribution of each module.

Components			MATH500		GSM-Plus		Avg	
Joint Train	Bias Estimator	Penalty	ArithACC(%)	Length	ArithACC(%)	Length	ArithACC(%)	Length
✓	✓	✓	31.4	309.2	<u>60.3</u>	<u>243.3</u>	45.9	276.3
✓	✓	×	<u>30.6</u>	387.7	60.4	273.1	<u>45.5</u>	330.4
×	✓	✓	31.0	<u>329.0</u>	59.9	238.3	45.5	<u>283.7</u>
×	✓	×	29.8	398.3	59.1	286.5	44.5	342.4
×	×	✓	30.0	376.1	59.3	280.5	44.7	328.3
×	×	×	30.4	411.1	59.1	287.9	44.8	349.5

Table 7: We assess the performance of CoLD PRM variants using Best-of-16 search with solutions sampled from the MetaMath-Mistral-7B model. Component-wise ablations are conducted to evaluate the contribution of each module.

Components			MATH500		GSM-Plus		Avg	
Joint Train	Bias Estimator	Penalty	ArithACC(%)	Length	ArithACC(%)	Length	ArithACC(%)	Length
✓	✓	✓	37.2	<u>376.3</u>	61.4	238.6	49.3	307.5
✓	✓	×	36.2	552.6	<u>61.3</u>	289.7	48.8	421.2
×	✓	✓	38.6	353.4	59.8	<u>262.7</u>	<u>49.2</u>	<u>308.1</u>
×	✓	×	<u>38.0</u>	554.1	58.9	322.8	48.5	438.4
×	×	✓	37.6	448.8	59.1	313.9	48.4	381.4
×	×	×	30.4	411.1	59.4	335.5	44.9	373.3

G Hyperparameter Study

G.1 Effect of c

We investigate the effect of the hyperparameter c , which controls the strength of the length bias correction in CoLD PRM. We evaluate the performance of CoLD PRM across different values of $c \in \{0.7, 0.9, 1.1, 1.3, 1.5\}$.

Figure 10 illustrates how varying c affects model accuracy and solution length. The optimal c varies when different models serve as policy models. For instance, when LLaMA-3-70B-Instruct acts as the policy model, it achieves the highest accuracy at $c = 1.1$; in contrast, when MetaMath-Mistral-7B functions as the policy model, it performs best at $c = 0.7$. Despite these discrepancies, a moderate value like $c = 1.1$ generally delivers strong results across all these policy models. This indicates that CoLD PRM is robust to the selection of c , and using a single c value can effectively balance accuracy and solution length without the need for extensive per-model tuning.

G.2 Effect of λ_b

We investigate the effect of the hyperparameter λ_b , which controls the strength of the correlation-based regularization in CoLD PRM. We evaluate the performance of CoLD PRM across different values of $\lambda_b \in \{0.3, 0.5, 0.7\}$.

Table 8: Performance of the CoLD framework under different noise scales k . Responses are sampled from the Llama-3-70B-Instruct model.

Noise Scale (k)	MATH500		GSM-Plus	
	ArithACC (%)	Length	ArithACC (%)	Length
0 (No noise)	47.2	337.9	71.4	238.2
5	49.2	313.2	73.8	202.5
10	48.9	308.6	73.1	200.7
15	48.8	301.1	73.5	210.8

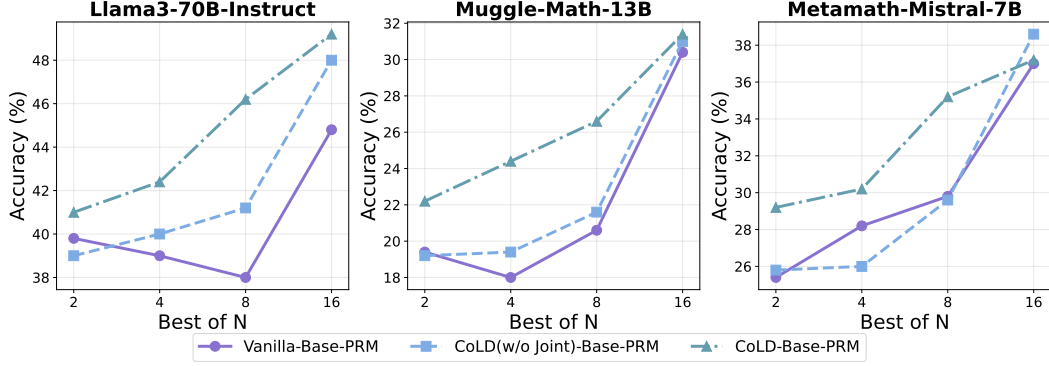


Figure 9: Performance of CoLD under different numbers of best-of-N samples on the Math dataset across different policy models.

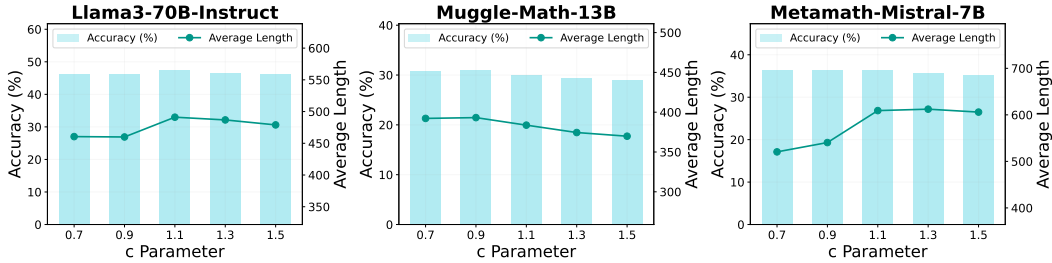


Figure 10: The performance of CoLD PRM under varying values of c across different policy models.

Figure 11 illustrates how varying the value of λ_b affects the model’s accuracy and the length of selected solutions. It is evident that different values of λ_b exhibit slight variations in performance across diverse datasets. Specifically, in certain cases, even when the length of the selected solution is relatively short, it may still result in a decline in accuracy. Consequently, an unwavering pursuit of brevity does not necessarily lead to the improvement of performance. Nevertheless, on the whole, CoLD PRM still demonstrates a certain degree of robustness.

G.3 Effect of α

To further analyze potential over-penalization, we conduct additional experiments on the parameter $\alpha \in \{0.1, 0.01, 0.001, 0.0001, 0.00001\}$ across three Policy Model. From the Table 9, we can see that within a reasonable range of α , CoLD demonstrates robustness, and the strength of the penalty controlled by α has very limited impact on performance. However, it is also clear that when α applies an excessively strong penalty, over-penalization occurs. This indicates that over-penalization can appear in certain extreme cases.

H Other Benchmark

To better evaluate the capabilities of our model, we assess it on ProcessBench [53]—a public benchmark for Process Reward Models (PRMs). The goal of this evaluation is to determine whether the PRM can identify the first erroneous step in the reasoning process. ProcessBench partitions the dataset into two subsets: one containing samples with incorrect final answers and the other with correct final answers. It then computes the harmonic mean of the accuracies achieved on these two subsets to obtain the final F1-score.

As shown in the Table 10, our CoLD PRM outperforms nearly all open-source PRM baselines across all datasets. The performance advantage is particularly notable on the more challenging datasets, including OlympiadBench and OmniMATH, demonstrating that our CoLD PRM maintains strong generalization over OOD datasets.

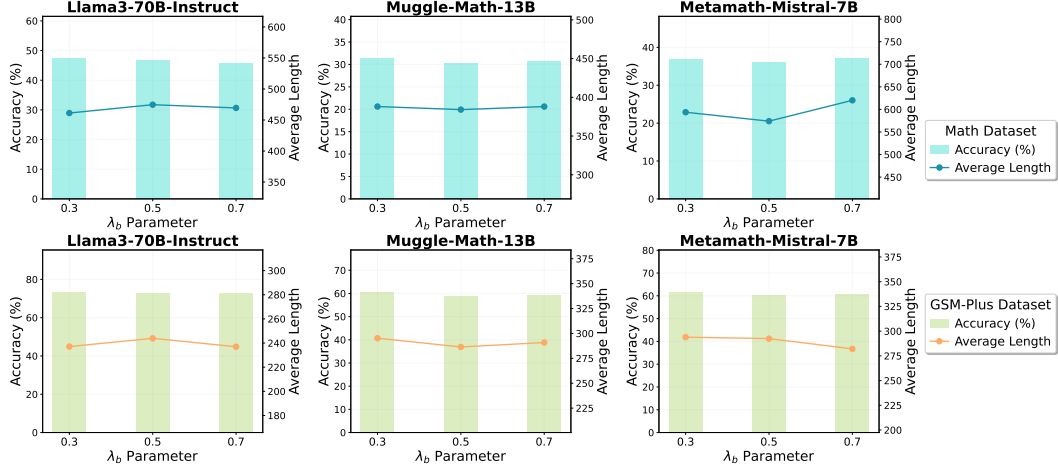


Figure 11: The performance of CoLD PRM under varying values of λ_b across different policy models.

Table 9: Effect of different α values on CoLD performance across policy models.

Policy Model	α	MATH500		GSM-Plus		Avg	
		ArithACC(%)	Length	ArithACC(%)	Length	ArithACC(%)	Length
Llama-3-70B-Instruct	0.1	38.8	222.6	68.6	177.4	53.7	200.0
	0.01	42.4	223.5	71.1	179.6	56.8	201.6
	0.001	48.6	253.5	72.6	189.7	60.6	221.6
	0.0001	49.2	313.2	73.8	202.5	61.5	257.9
	0.00001	47.9	350.6	73.3	210.4	60.6	280.5
MetaMath-Mistral-7B	0.1	25.0	215.3	45.5	202.9	35.3	209.1
	0.01	31.2	218.9	50.2	205.3	40.7	212.1
	0.001	38.8	276.29	56.9	220.2	47.9	248.2
	0.0001	37.2	376.3	61.4	238.6	49.3	307.5
	0.00001	36.2	393.8	59.9	259.4	48.1	326.6
Muggle-Math-13B	0.1	21.4	191.1	43.7	193.1	32.6	192.1
	0.01	27.0	195.9	50.0	195.8	38.5	195.9
	0.001	30.2	248.4	58.5	215.8	44.4	232.1
	0.0001	31.4	309.2	60.3	243.3	45.9	276.3
	0.00001	30.2	357.7	59.62	263.2	44.9	310.4

When comparing models across different scales, CoLD still maintains strong competitiveness. In terms of overall performance, it is even able to outperform some larger-scale counterpart models.

I Evaluation Using Stronger Policy Model

In addition to the policy models reported in the main paper, we further include Qwen3-Next-80B-A3B-Instruct [42] and DeepSeek-v3.1 [20] in our evaluation. These models represent state-of-the-art large-scale language models that have demonstrated substantially stronger reasoning capabilities compared with the earlier baselines.

As shown in the Table 11, our CoLD framework continues to deliver clear and consistent improvements even when applied to these more advanced LLMs. This confirms that CoLD is not merely compensating for weaknesses in smaller models, but instead provides generalizable gains that hold across diverse policy models, including the latest high-performing LLMs.

J Bias Estimator Interpretability

To better understand what the Bias Estimator learns and how it makes predictions, we conduct interpretability analyses focusing on its sensitivity to input length versus semantic content. Through a combination of output case studies and attention-based comparisons, we aim to clarify whether the

Table 10: The performance of different models on ProcessBench. The best result is given in **bold**, and the second-best value is underlined.

Model		GSM8k		MATH		OlympiadBench		OmniMATH		Avg.F1
		ArithACC	F1	ArithACC	F1	ArithACC	F1	ArithACC	F1	
Open-source PRM	CoLD-Base-PRM-7B(Ours)	<u>72.0</u>	68.6	67.3	67.7	54.6	56.0	47.8	51.3	60.9
	Qwen2.5-Math-7B-PRM800K	73.5	68.2	<u>65.1</u>	<u>62.6</u>	<u>53.2</u>	<u>50.7</u>	<u>43.4</u>	<u>44.3</u>	<u>56.5</u>
	Skywork-PRM-7B	71.6	70.8	54.5	53.6	25.6	22.9	23.7	21.0	42.1
	Skywork-PRM-1.5B	59.9	59.0	49.1	48.0	20.5	19.3	19.7	19.2	36.4
	Math-Shepherd-PRM-7B	58.3	47.9	45.1	29.5	39.7	24.8	34.8	23.8	31.5
	RLHFlow-PRM-Mistral-8B	62.3	50.4	42.1	33.4	22.3	13.8	19.1	15.8	28.4
	RLHFlow-PRM-Deepseek-8B	56.9	38.8	45.1	33.8	26.5	16.9	23.2	16.9	26.6
Language Models as Critic	QwQ-32B-Preview	87.9	88.0	78.5	78.7	<u>59.2</u>	57.8	61.1	61.3	71.5
	GPT-4o	80.2	79.2	63.4	<u>63.6</u>	50.1	51.4	50.1	<u>53.5</u>	<u>61.9</u>
	Qwen2.5-72B-Instruct	77.9	76.2	<u>65.4</u>	61.8	59.8	<u>54.6</u>	55.1	52.2	61.2
	Llama-3.3-70B-Instruct	<u>83.7</u>	<u>82.9</u>	63.7	59.4	54.3	46.7	51.0	43.0	58.0
	Qwen2.5-Coder-32B-Instruct	72.0	68.9	64.5	60.1	57.0	48.9	52.5	46.3	56.1
	Llama-3.1-70B-Instruct	75.3	74.9	52.6	48.2	50.0	46.7	43.2	41.0	52.7
	Qwen2.5-14B-Instruct	72.3	69.3	59.2	53.3	50.2	45.0	43.5	41.3	52.2
	Qwen2-72B-Instruct	67.8	67.6	52.3	49.2	43.3	42.1	39.3	40.2	49.8
	Qwen2.5-32B-Instruct	70.6	65.6	61.9	53.1	53.5	40.0	47.7	38.3	49.3
	Qwen2.5-Math-72B-Instruct	70.3	65.8	59.6	52.1	56.1	32.5	55.1	31.7	45.5
	Qwen2.5-Coder-14B-Instruct	61.9	50.1	54.2	39.9	51.4	34.0	<u>55.6</u>	27.3	37.8
	Qwen2.5-7B-Instruct	37.8	36.5	36.9	36.6	29.9	29.7	27.3	27.4	32.6
	Meta-Llama-3-70B-Instruct	62.4	52.2	48.3	22.8	46.2	21.2	44.8	20.0	29.1
	Qwen2.5-Math-7B-Instruct	54.4	26.8	50.3	25.7	43.1	14.2	41.6	12.7	19.9
	Qwen2-7B-Instruct	25.1	8.4	20.4	19.0	16.1	14.7	13.8	12.1	13.6
	Meta-Llama-3-8B-Instruct	27.1	13.1	17.3	13.8	14.2	4.8	19.7	12.6	11.1
	Qwen2.5-Coder-7B-Instruct	49.1	14.3	46.3	6.5	47.2	4.1	48.9	1.8	6.7
	Llama-3.1-8B-Instruct	27.3	10.9	20.5	5.1	16.0	2.8	15.0	1.6	5.1

Table 11: Evaluation Results of CoLD Using Advanced Large-Scale Policy Models on MATH500.

Policy Model	PRM model	ArithACC(%)	Length
Qwen3-Next-80B -A3B-Instruct	Vanilla Base PRM	85.2	1050.6
	CoLD Base PRM(Ours)	90.4	754.5
Deepseek-v3.1	Vanilla Base PRM	89.6	953.9
	CoLD Base PRM(Ours)	91.6	646.2

Bias Estimator captures spurious surface-level cues, specifically length, rather than the underlying meaning of the input.

J.1 Case Study

To illustrate the interpretability of the Bias Estimator, we present a case study demonstrating its output behavior. As shown in Figure 12, the Bias Estimator responds more strongly to length than to semantic content. For example, in Step 3 of the two answers displayed, although their semantic meaning differs and one step is correct while the other is incorrect, the Bias Estimator assigns similar scores because it is primarily driven by step length rather than semantics.

J.2 Attention Visualization

- Sentence A: Compute r . Here $x = 0$ and $y = 3$, so $r = \sqrt{0^2 + 3^2} = \sqrt{9} = 3$.
- Sentence B: Analyze k . If a machine runs 12 cycles in 4 seconds, then $k = 12/4 = 3$.

To examine whether the Bias Estimator relies on length rather than semantic information, we conduct two comparative attention analyses. First, we feed Sentence A into both the Bias Estimator and the backbone model and compare their attention distributions. As shown in Figure 13, while the backbone exhibits semantically focused attention peaks, the Bias Estimator assigns attention almost uniformly across tokens, without concentrating on semantic-bearing words. Second, we input Sentence A and Sentence B, which share the same length but differ completely in semantics, into the Bias Estimator. As illustrated in Figure 14, their attention maps remain nearly indistinguishable, indicating that the attention pattern remains stable when semantics change while length is kept constant. Together, these results reinforce that the Bias Estimator’s attention primarily encodes length-related cues rather than semantic information.

Question: Convert the point $(0,3)$ in rectangular coordinates to polar coordinates. Enter your answer in the form (r,θ), where $r > 0$ and $0 \leq \theta < 2\pi$.
Answer1: Step 1. Use the formulas (r,θ) satisfy $r = \sqrt{x^2 + y^2}$, $\theta = \operatorname{atan2}(y,x)$ (0.2806) Step 2. Compute r. Here $x=0$ and $y=3$, so $r = \sqrt{0^2 + 3^2} = \sqrt{9} = 3$. (0.3257) Step 3. Determine θ. The point $(0,3)$ is on the positive y-axis, so $\theta = \frac{\pi}{2}$. (0.3541) Step 4. State the final answer: $(0,3) \rightarrow (3, \frac{\pi}{2})$. (0.2556)
Answer2: Step 1. Use the formulas (r,θ) satisfy $r = \sqrt{x^2 + y^2}$, $\theta = \operatorname{atan2}(y,x)$ (0.2806) Step 2. Compute r. Here $x=0$ and $y=3$, so $r = \sqrt{0^2 + 3^2} = \sqrt{9} = 3$. (0.3257) Step 3. Determine θ. Using $\theta = \arctan\left(\frac{y}{x}\right)$ gives $\theta = \arctan\left(\frac{3}{0}\right) = 0$. (0.3491) Step 4. State the final answer: $(0,3) \rightarrow (3, 0)$. (0.2456)

Figure 12: Case study of the Bias Estimator outputs, where the red numbers in parentheses at the end of each step denote the predicted scores.

K Limitation

We identify two primary limitations in our approach. First, the framework relies on heuristic noise injection to approximate counterfactual scenarios. A true counterfactual requires a reasoning step strictly stripped of all semantic content while retaining its exact surface length. Since this ideal state is fundamentally unobservable, the Bias Estimator depends on perturbation approximations. Consequently, the estimated causal effect of length remains a proxy, and the bias estimation could potentially be influenced by residual semantic leakage depending on the noise configuration. Second, the framework introduces additional hyperparameters, specifically the bias correction strength (c), the correlation regularization weight (λ_b), and the length penalty (α). The presence of these variables intrinsically increases the search space for hyperparameter tuning. This expanded search space could complicate deployment and scaling across highly diverse or unseen policy models in practical applications.

L Broader Impacts

Our research addresses the phenomenon of length bias in Process Reward Models (PRMs), yielding several positive societal impacts. By explicitly disentangling true semantic correctness from superficial verbosity, our CoLD framework promotes fairer and more reliable automated evaluation for Large Language Models (LLMs). This mitigates the risk of reward hacking, where models are incentivized to generate redundant or hallucinatory text simply to attain higher scores. Additionally, by encouraging more concise and precise reasoning trajectories, our approach can help reduce the computational overhead and carbon footprint associated with LLM inference, which aligns with the objectives of environmentally sustainable AI.

Regarding potential negative impacts, our work shares the dual-use risks inherent to any foundational advancement in AI reasoning. While our framework is designed to improve logical correctness in mathematical problem-solving, enhanced reward modeling techniques could theoretically be

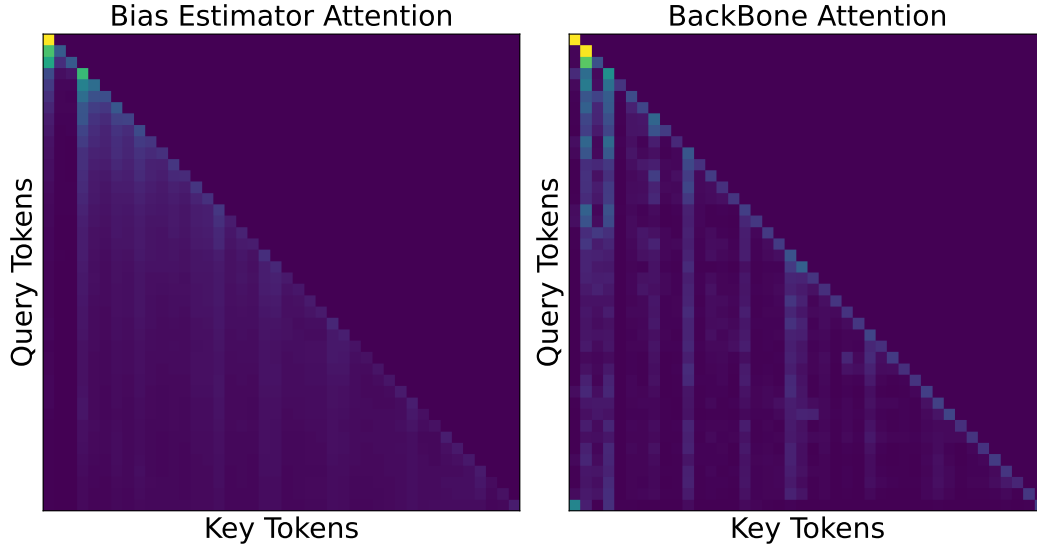


Figure 13: Attention visualizations for Sentence A in the backbone model and the Bias Estimator.

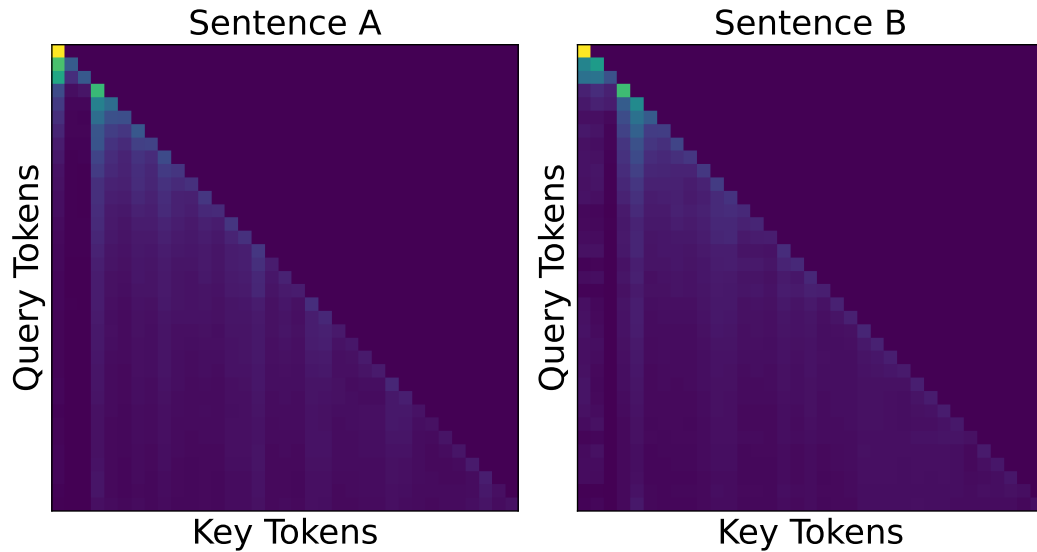


Figure 14: Attention visualizations for Sentence A and Sentence B, which share identical length but differ in semantics, under the Bias Estimator.

repurposed by bad actors to optimize models for generating sophisticated deceptive logic or malicious code. Since our focus is on foundational methodology, mitigating these downstream risks relies on the broader AI community to implement robust safety guardrails and usage monitoring at the final deployment stage.

Large Language Models (LLMs) Usage

In this work, large language models (LLMs) were used solely for text polishing and language refinement. They were not involved in the design of the methodology, implementation, analysis, or the generation of experimental results. All technical contributions and research findings are entirely the work of the authors.

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